Joining data in real life

PANDAS JOINS FOR SPREADSHEET USERS

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Mixing columns and indexes

index	player_id	name	team
0	3897	Barry Church	
1	18208	Sheldon Rankins	New Orleans Saints
2	21284	Grover Stewart	Indianapolis Colts
3	20013	Tyler Shatley	
4	8204	Brandon	
5	7636	Jimmy Ga	

player_id	name	position
10	Ameer Abdullah	RB
40	Sam Acho	OLB
42	Kenneth Acker	CB
50	Andrew Adams	S
60	Davante Adams	WR
68	Jamal Adams	SS

```
teams.merge(positions, left_on='player_id', right_index=True)
```

or

```
positions.merge(teams, left_index=True, right_on='player_id')
```

Working with overlapping column names

current	name	team
0	Charles Sims	Tampa Bay Buccaneers
1	Marwin Evans	Green Bay Packers
2	Philip R	
3	Chris Man	
4	Ryan Sw	

drafted	name	team
0	Barry Church	None
1	Sheldon Rankins	New Orleans Saints
2	Grover Stewart	Indianapolis Colts
3	Tyler Shatley	None
4	Brandon Graham	Philadelphia Eagles

	name	team_current	team_drafted
0	Marwin Evans	Green Bay Packers	None
1	Philip Rivers	Los Angeles Chargers	New York Giants
2	Chris Manhertz	Carolina Panthers	None
3	Ryan Switzer	Dallas Cowboys	Dallas Cowboys
4	Terrance Williams	Dallas Cowboys	Dallas Cowboys

```
current.merge(drafted, on='name', suffixes=('_current', '_drafted'))
```

Identifying rows by table

	name	team
0	Tyrann Mathieu	Arizona Cardinals
1	Drew Stanto	
2	Antoine Bethe	
3	David Johnso	

	name	team
0	John Brown	Arizona Cardinals
1	Larry Fitzgerald	Arizona Cardinals
2	David Johnson	Arizona Cardinals
3	Evan Boehm	Arizona Cardinals

	name	team_current	team_drafted	_merge
0	Tyrann Mathieu	Arizona Cardinals	NaN	left_only
1	Drew Stanton	Arizona Cardinals	NaN	left_only
2	Antoine Bethea	Arizona Cardinals	NaN	left_only
3	David Johnson	Arizona Cardinals	Arizona Cardinals	both
4	John Brown	NaN	Arizona Cardinals	right_only
5	Larry Fitzgerald	NaN	Arizona Cardinals	right_only
6	Evan Boehm	NaN	Arizona Cardinals	right_only

```
current.merge(drafted, how='outer', on='name', suffixes=('_current', '_drafted'),
              indicator=True)
```

Sorting rows by key

	name	team_current	team_drafted
0	A.J. Bouye	NaN	None
1	A.J. Cann	NaN	Jacksonville Jaguars
2	A.J. Derby	NaN	New England Patriots
3	A.J. Francis	Washington Redskins	None
4	A.J. Green	Cincinnati Bengals	Cincinnati Bengals

```
current.merge(drafted, on='name', suffixes=('_current', '_drafted'),
              sort=True)
```

Let's practice!

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Working with time data

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Joining with .merge_ordered()

CLEVELAND

	Game_Date	Temperature
0	2016-08-18	79
1	2016-09-01	73
2	2016-09-18	[illegible]
3	2016-10-09	[illegible]
4	2016-10-30	[illegible]

DALLAS

	Game_Date	Temperature
0	2016-08-19	82
1	2016-09-01	90
2	2016-09-11	85
3	2016-09-25	86
4	2016-10-09	80

```
pd.merge_ordered(cleveland, dallas, on='Game_Date',
                 suffixes=['_CLE', '_DAL'])
```

Joining with .merge_ordered()

CLEVELAND

	Game_Date	Temperature
0	2016-08-18	79
1	2016-09-01	73
2	2016-09-18	
3	2016-10-09	
4	2016-10-30	

DALLAS

	Game_Date	Temperature
0	2016-08-19	82
1	2016-09-01	90
2	2016-09-11	85
3	2016-09-25	86
4	2016-10-09	80

	Game_Date	Temperature_CLE	Temperature_DAL
0	2016-08-18	79.0	NaN
1	2016-08-19	NaN	82.0
2	2016-09-01	73.0	90.0
3	2016-09-11	NaN	85.0
4	2016-09-18	74.0	NaN
5	2016-09-25	NaN	86.0
6	2016-10-09	58.0	80.0
7	2016-10-30	53.0	84.0

```
pd.merge_ordered(cleveland, dallas, on='Game_Date',
                 suffixes=['_CLE', '_DAL'])
```

Interpolating data

CLEVELAND

	Game_Date	Temperature
0	2016-08-18	79
1	2016-09-01	73
2	2016-09-18	
3	2016-10-09	
4	2016-10-30	

DALLAS

	Game_Date	Temperature
0	2016-08-19	82
1	2016-09-01	90
2	2016-09-11	85
3	2016-09-25	86
4	2016-10-09	80

	Game_Date	Temperature_CLE	Temperature_DAL
0	2016-08-18	79	NaN
1	2016-08-19	79	82.0
2	2016-09-01	73	90.0
3	2016-09-11	73	85.0
4	2016-09-18	74	85.0
5	2016-09-25	74	86.0
6	2016-10-09	58	80.0
7	2016-10-30	53	84.0

```
pd.merge_ordered(tc2, td2, on='Game_Date',
                 suffixes=['_CLE', '_DAL'], fill_method='ffill')
```

Interpolating data

CLEVELAND

	Game_Date	Temperature
0	2016-08-18	79
1	2016-09-01	73
2	2016-09-18	
3	2016-10-09	
4	2016-10-30	

DALLAS

	Game_Date	Temperature
0	2016-08-19	82
1	2016-09-01	90
2	2016-09-11	85
3	2016-09-25	86
4	2016-10-09	80

	Game_Date	Temperature_CLE	Temperature_DAL
0	2016-08-18	79	NaN
1	2016-08-19	79	82.0
2	2016-09-01	73	90.0
3	2016-09-11	73	85.0
4	2016-09-18	74	85.0
5	2016-09-25	74	86.0
6	2016-10-09	58	80.0
7	2016-10-30	53	84.0

```
pd.merge_ordered(tc2, td2, on='Game_Date',
                 suffixes=['_CLE', '_DAL'], fill_method='ffill')
```

Merging to nearest date-times

- pandas.merge_asof()
- matches on nearest date
- similar to VLOOKUP(range_lookup=TRUE)

```
pd.merge_asof(left_df, right_df,
              direction='backward')
```

Directions

- "backward": closest date that is earlier
- "forward": closest date equal or later
- "nearest": closest date regardless

Merge_asof Example

	Game_Date	Temperature_cle	Temperature_dal
0	2016-08-18	79	NaN
1	2016-08-19	79	82.0
2	2016-09-01		
3	2016-09-11		
4	2016-09-18		

	impact_count	cume_count
Game_Date		
2016-08-11	1	1
2016-08-19	1	2
2016-08-20	1	3
2016-08-27	1	4

	Game_Date	Temperature_cle	Temperature_dal	cume_count
0	2016-08-18	79	NaN	1
1	2016-08-19	79	82.0	2
2	2016-09-01	73	90.0	6
3	2016-09-11	73	85.0	6
4	2016-09-18	74	85.0	6
5	2016-09-25	74	86.0	6
6	2016-10-09	58	80.0	6
7	2016-10-30	53	84.0	8

```
pd.merge_asof(temps, impacts,
              left_on='Game_Date', right_index=True, direction='backward')
```

Let's practice!

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Recap and case study

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Chapter 1 - Introduction to joining data

- Common situations
- Concatenate data by row or column

```
pd.concat([df1, df2], axis=0)
```

- Pandas as a powerful tool

Chapter 2 - VLOOKUP-style joins

One-to-one, VLOOKUP-style joins

```
pd.merge(left_df, right_df,
         how='inner',
         on='key_column')
```

GameKey	Season
2	2016
4	2016
6	2016
8	2016

GameKey	Turf	Stadium
1	Turf	Tom Benson Hall of Fame
2	Grass	Los Angeles Memorial Col
3	Natural Grass	M&T Bank Stadium
4	DD GrassMaster	Lambeau Field
5	Grass	Soldier Field
6	Grass	Heinz Field
7	Natural grass	Levis Stadium
8	A-Turf Titan	Ralph Wilson Stadium

GameKey	Season	Turf	Stadium
2	2016	Grass	Los Angeles Memorial
4	2016	DD GrassMaster	Lambeau Field
6	2016	Grass	Heinz Field
8	2016	A-Turf Titan	Ralph Wilson Stadium

Chapter 3- One-to-many joins

Joins (merges) on key column

```
df1.merge(df2, how='inner',
          on='key_column')
```

Joins on unique index

```
df1.join(df2, how='left')
```

	A	B	C
1	GameKey	Season	SeasonType
2	2	2016	Pre
3	3	2016	Pre
4	4	2016	Pre
5	5	2016	Pre
6	6	2016	Pre
7	7	2016	Pre
8	8	2016	Pre
9	9	2016	Pre

	A	B	C
1	GameKey	PlayId	GameClock
2	2	191	12:30
3	2	1132	12:08
4	3	1676	1:51
5	3	2643	6:10
6	3	3949	1:59
7	4	291	9:32
8	4	386	8:37
9	4	927	1:53

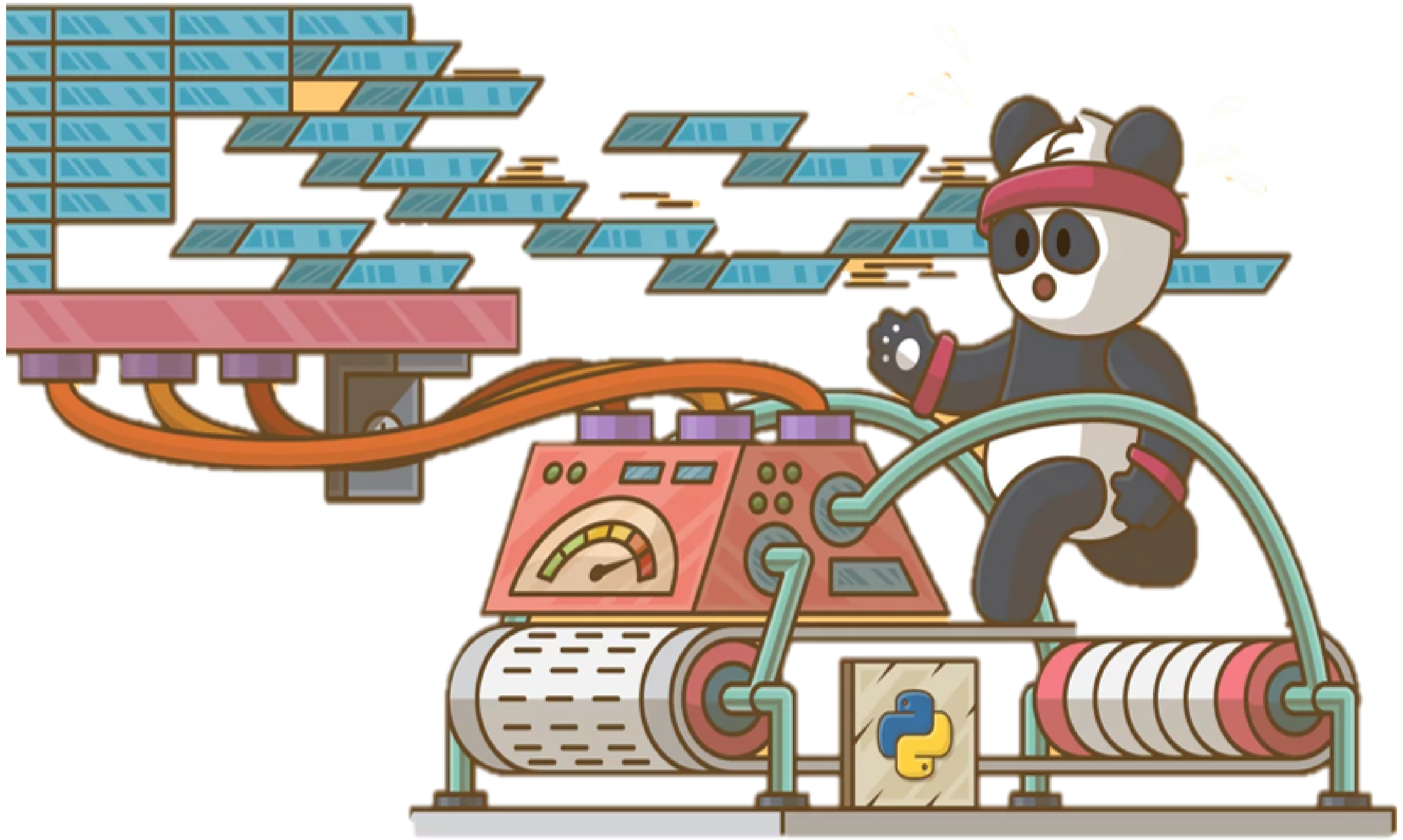
	A	B	C	D	E
1	GameKey	Season	SeasonType	PlayId	GameClock
2	2	2016	Pre	191	12:30
3	2	2016	Pre	1132	12:08
4	3	2016	Pre	1676	1:51
5	3	2016	Pre	2643	6:10
6	3	2016	Pre	3949	1:59
7	4	2016	Pre	291	9:32
8	4	2016	Pre	386	8:37
9	4	2016	Pre	927	1:53

Chapter 4 - Advanced joins

Advanced parameters

- left_index, right_index
- suffixes
- indicator
- sort

```
pd.merge_ordered(left_df, right_df,
                 how='outer')
pd.merge_asof(left_df, right_df,
              direction='backward')
```



Thank you!

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